```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

sns.set_style("whitegrid")
```

```
df = pd.read_csv("Titanic-Dataset.csv")

df.head()
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	NaN	S
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833	C85	C
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250	NaN	S
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000	C123	S
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.0500	NaN	S

Next steps: Generate code with df New interactive sheet

```
print(df.shape)

df.info()

df.describe()
```

```
(891, 12)
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
 #   Column       Non-Null Count  Dtype  
---  ------       --------------  -----  
 0   PassengerId  891 non-null    int64  
 1   Survived     891 non-null    int64  
 2   Pclass       891 non-null    int64  
 3   Name         891 non-null    object 
 4   Sex          891 non-null    object 
 5   Age          714 non-null    float64
 6   SibSp        891 non-null    int64  
 7   Parch        891 non-null    int64  
 8   Ticket       891 non-null    object 
 9   Fare         891 non-null    float64
 10  Cabin        204 non-null    object 
 11  Embarked     889 non-null    object 
dtypes: float64(2), int64(5), object(5)
memory usage: 83.7+ KB
```

	PassengerId	Survived	Pclass	Age	SibSp	Parch	Fare
count	891.000000	891.000000	891.000000	714.000000	891.000000	891.000000	891.000000
mean	446.000000	0.383838	2.308642	29.699118	0.523008	0.381594	32.204208
std	257.353842	0.486592	0.836071	14.526497	1.102743	0.806057	49.693429
min	1.000000	0.000000	1.000000	0.420000	0.000000	0.000000	0.000000
25%	223.500000	0.000000	2.000000	20.125000	0.000000	0.000000	7.910400
50%	446.000000	0.000000	3.000000	28.000000	0.000000	0.000000	14.454200
75%	668.500000	1.000000	3.000000	38.000000	1.000000	0.000000	31.000000
max	891.000000	1.000000	3.000000	80.000000	8.000000	6.000000	512.329200

```
df.isnull().sum()
```

	0
PassengerId	0
Survived	0
Pclass	0
Name	0
Sex	0
Age	177
SibSp	0
Parch	0
Ticket	0
Fare	0
Cabin	687
Embarked	2

dtype: int64

```
df["Age"] = df["Age"].fillna(df["Age"].mean())
```

```
df["Age"] = df["Age"].fillna(df["Age"].mean())
```

```
df["Embarked"] = df["Embarked"].fillna(df["Embarked"].mode()[0])
```

```
df.drop("Cabin", axis=1, inplace=True)
```

```
df.isnull().sum()
```

	0
PassengerId	0
Survived	0
Pclass	0
Name	0
Sex	0
Age	0
SibSp	0
Parch	0
Ticket	0
Fare	0
Embarked	0

dtype: int64

```
df["FamilySize"] = df["SibSp"] + df["Parch"]
```

```
df.head()
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Embarked	Famil
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	S	
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833	C	
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250	S	
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000	S	
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.0500	S	

Next steps: Generate code with df New interactive sheet

```
bins = [0, 12, 19, 35, 60, 100]

labels = [
    "Child",
    "Teen",
    "Young Adult",
    "Adult",
    "Senior"
]

df["AgeGroup"] = pd.cut(
    df["Age"],
    bins=bins,
    labels=labels
)
```

```
age_survival = (
    df.groupby("AgeGroup")["Survived"]
    .mean()
    .sort_values(ascending=False)
)

print(age_survival)
```

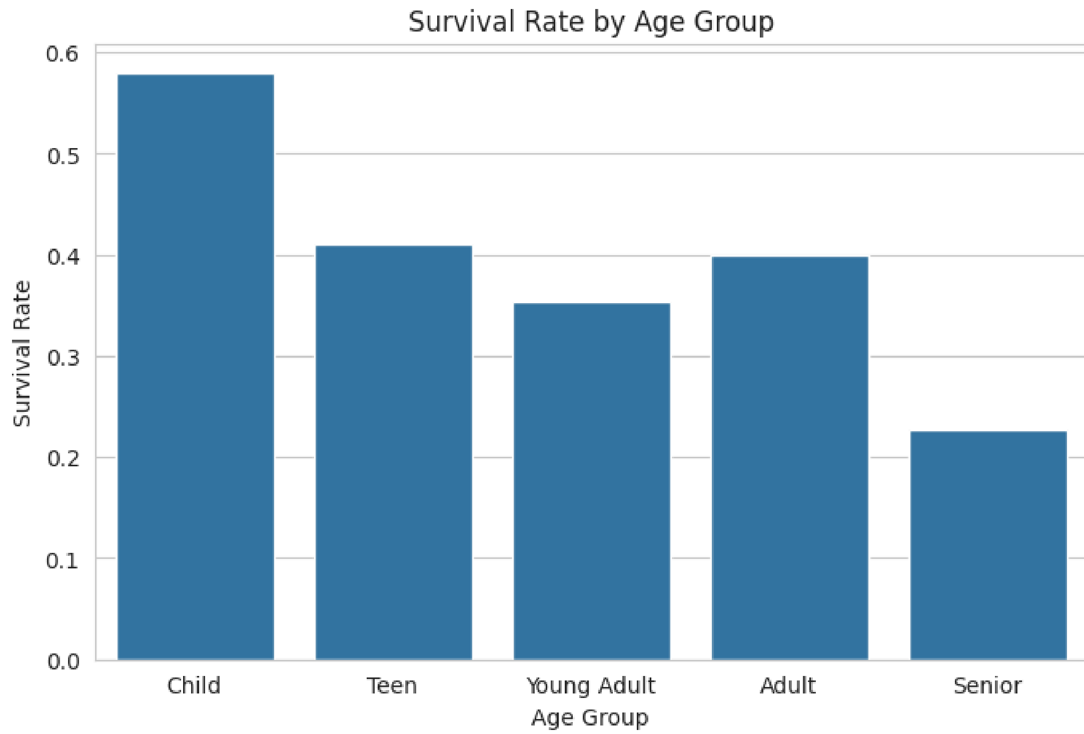
```
AgeGroup
Child          0.579710
Teen           0.410526
Adult          0.400000
Young Adult    0.352941
Senior         0.227273
Name: Survived, dtype: float64
/tmp/ipykernel_1278/1035498245.py:2: FutureWarning: The default of observed=False is deprecated and will
  df.groupby("AgeGroup")["Survived"]
```

```
plt.figure(figsize=(8,5))

sns.barplot(
    x=age_survival.index,
    y=age_survival.values
)

plt.title("Survival Rate by Age Group")
plt.xlabel("Age Group")
plt.ylabel("Survival Rate")

plt.show()
```




```
embark_survival = (
    df.groupby("Embarked")["Survived"]
    .mean()
)

print(embark_survival)
```

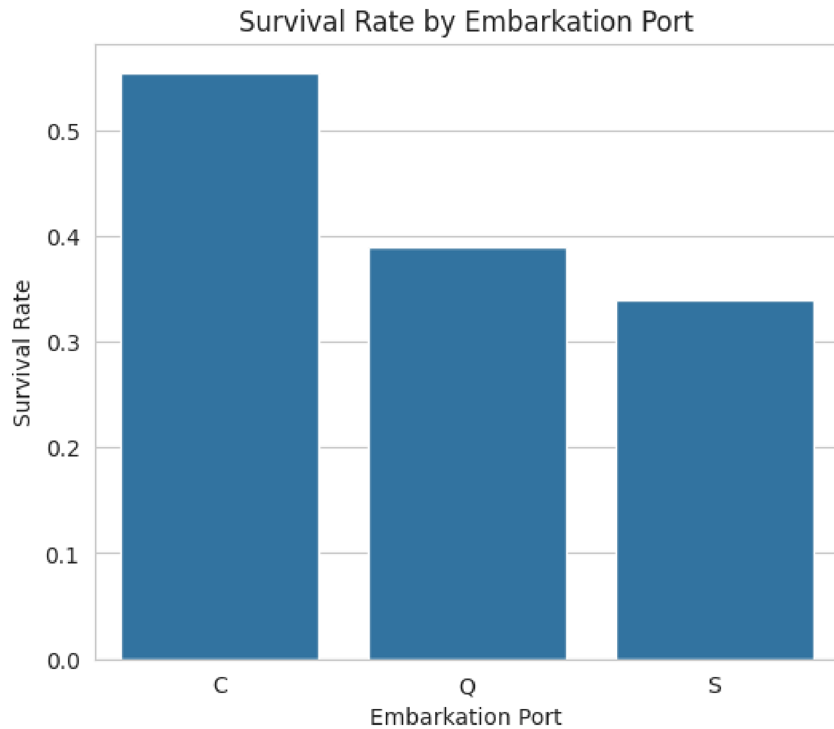
```
Embarked
C    0.553571
Q    0.389610
S    0.339009
Name: Survived, dtype: float64
```

```
plt.figure(figsize=(6,5))

sns.barplot(
    x=embark_survival.index,
    y=embark_survival.values
)

plt.title("Survival Rate by Embarkation Port")
plt.xlabel("Embarkation Port")
plt.ylabel("Survival Rate")

plt.show()
```




```
family_survival = (
    df.groupby("FamilySize")["Survived"]
    .mean()
)

print(family_survival)
```

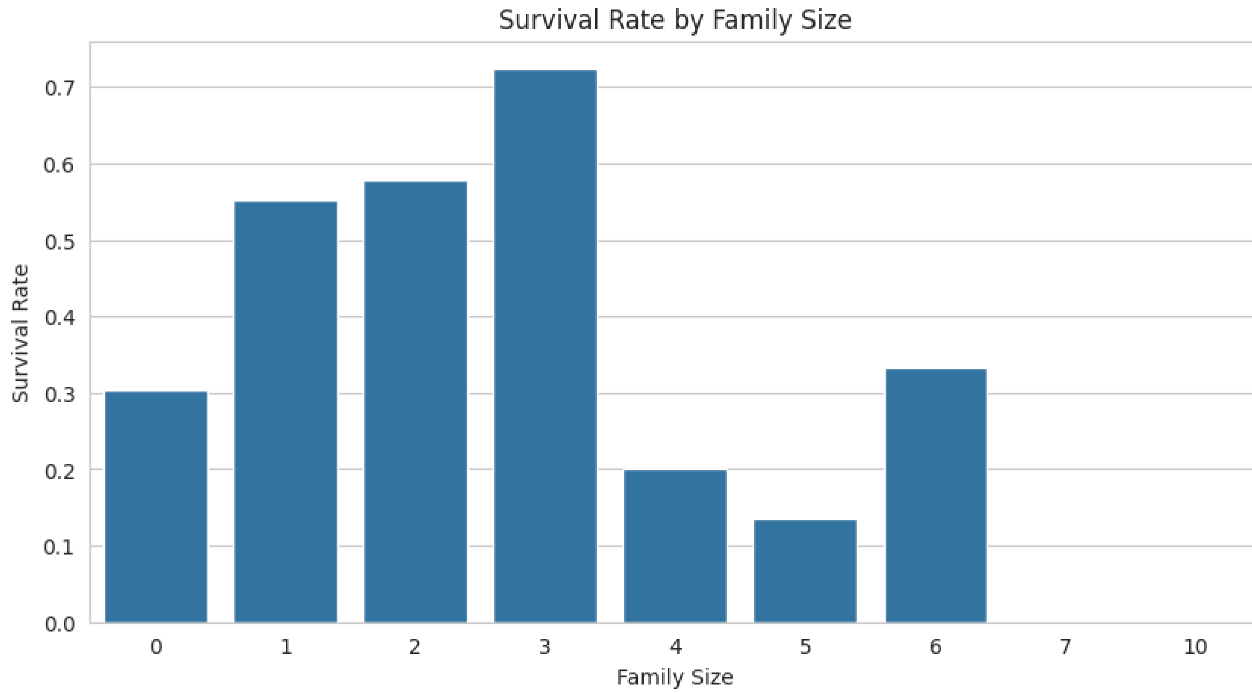
```
FamilySize
0     0.303538
1     0.552795
2     0.578431
3     0.724138
4     0.200000
5     0.136364
6     0.333333
7     0.000000
10    0.000000
Name: Survived, dtype: float64
```

```
plt.figure(figsize=(10,5))

sns.barplot(
    x=family_survival.index,
    y=family_survival.values
)

plt.title("Survival Rate by Family Size")
plt.xlabel("Family Size")
plt.ylabel("Survival Rate")

plt.show()
```

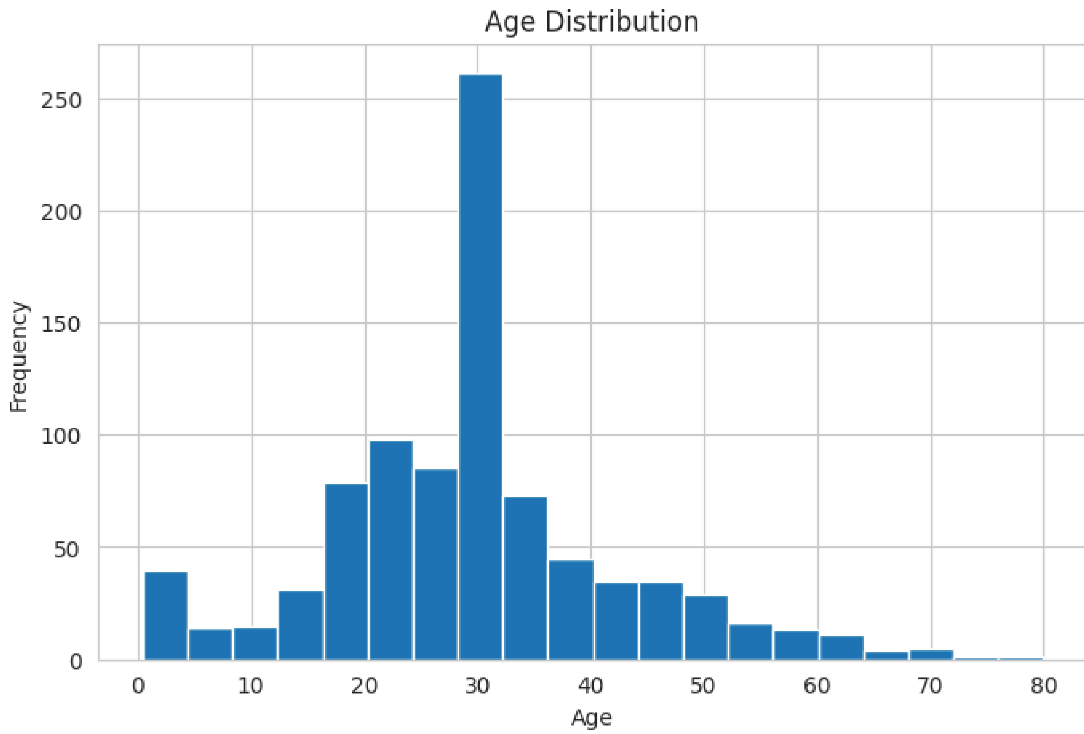



```
plt.figure(figsize=(8,5))

plt.hist(
    df["Age"],
    bins=20
)

plt.title("Age Distribution")
plt.xlabel("Age")
plt.ylabel("Frequency")

plt.show()
```




```
plt.figure(figsize=(10,6))

numeric_df = df.select_dtypes(include=np.number)

sns.heatmap(
    numeric_df.corr(),
    annot=True,
    cmap="coolwarm"
)

plt.title("Correlation Heatmap")

plt.show()
```